

PHYS 220 Introduction to Computational Physics

Dates and Location:

Tuesday & Thursday 8:30PM - 9:45PM

Policy Building 206

INSTRUCTOR: Dr. Zhibo Zhang

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OFFICE HOURS: PHYS417: Friday 11:00AM~12:00PM or Through Email

Recommended TEXTS:

“A Survey of Computational Physics: *introductory computational science*” By R. H. Landau et al.

Python Multimodal eTextBook publically available from

<http://www.compadre.org/psrc/items/detail.cfm?ID=11578>

GRADING:

Homework (30%), Midterm Project (20%), Final Project (30%),
Participation/Discussion (20%)

COURSE OUTLINE:

❖ Computer Setup And Programing Warm-Up

- Computer Setup
- Python programming basics
- Numpy and Matplotlib

❖ Understanding Errors and Uncertainties In Numerical Computations

- Type of Errors
- Tricks to control errors

❖ Monte Carlo method

- 2-D random walk
- 3-D random walk
- Real-world Problem: Photon scattering in cloud

❖ Numerical Integration

- Quadrature methods

- Monte Carlo method
- Real-world Problem: Integrate Radiance to Flux

❖ **Numerical Differentiation and Root Searching**

- Bisection method
- Newton-Raphson method
- Real-world Problem: Cloud property remote sensing

❖ **Midterm Projects**

❖ **Linear algebra and matrix computing**

- Matrix inversion and Eigenvalue
- Singular value decomposition
- Real-world problem: Greenhouse effect and atmospheric temperature profile

❖ **Data fitting**

- Quadrature fitting
- Least-square fitting
- Real-world problem: Satellite data analysis

❖ **Differential Equations: ODE and PDE (if time permits)**

- Trajectory
- Wave equation
- Real-world problem: Maxwell Eq. and E&M wave propagation

❖ **Final Projects**

Academic Integrity By enrolling in this course, each student assumes the responsibilities of an active participant in UMBC's scholarly community in which everyone's academic work and behavior are held to the highest standards of honesty. Cheating, fabrication, plagiarism, and helping others to commit these acts are all forms of academic dishonesty, and they are wrong. Academic misconduct could result in disciplinary action that may include, but is not limited to, suspension or dismissal. To read the full Student Academic Conduct Policy, consult the UMBC Student Handbook, the Faculty Handbook, or the UMBC Policies section of the UMBC Director.

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